

# Submission Worksheet

## Submission Data

**Course:** IT202-450-M2025

**Assignment:** IT202 PHP Multi-Dimension Problems

**Student:** Colin R. (ctr26)

**Status:** Submitted | **Worksheet Progress:** 100%

**Potential Grade:** 1000.00/1000.00 (100.00%)

**Received Grade:** 0.00/1000.00 (0.00%)

**Started:** 7/13/2025 10:02:51 PM

**Updated:** 7/13/2025 10:29:11 PM

**Grading Link:** <https://learn.ethereallab.app/assignment/v3/IT202-450-M2025/it202-php-multi-dimension-problems/grading/ctr26>

**View Link:** <https://learn.ethereallab.app/assignment/v3/IT202-450-M2025/it202-php-multi-dimension-problems/view/ctr26>

## Instructions

- Overview Link: <https://youtu.be/jd5giyXReal>
- 1. Ensure you read all instructions and objectives before starting.
- 2. Create a new branch from dev called M6-Homework
  - 1. `git checkout dev` (ensure proper starting branch)
  - 2. `git pull origin dev` (ensure history is up to date)
  - 3. `git checkout -b M6-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M6 Homework](#)
  - It includes Problems 1-3 and `base.php`. Put all into an M6 folder or similar inside your `public_html`
  - Immediately record to history
    - `git add public_html`
    - `git commit -m "adding M6 HW baseline files"`
    - `git push origin M6-Homework`
    - Create a Pull Request from M6-Homework to dev and keep it open
- 4. Fill out the below worksheet
  - Each Problem requires the following as you work
    - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
    - Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
    - Code solution (add/commit periodically as needed)
- 5. Once finished, click "Submit and Export"
- 6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
  - 1. `git add .`
  - 2. `git commit -m "adding PDF"`
  - 3. `git push origin M6-Homework`
  - 4. On Github merge the pull request from M6-Homework to dev
  - 5. On Github create a pull request from dev to prod and immediately merge. (This will trigger the prod deploy to make the heroku prod links work)

7. Upload the same PDF to Canvas
8. Sync Local
  1. git checkout dev
  2. git pull origin dev

## Section #1: ( 250 pts.) Problem 1 - Subset

Progress: 100%

### ≡ Task #1 ( 250 pts.) - Edit the `processBirds` function to extract properties

Progress: 100%

#### Details:

- Only make edits where noted via provided comments
- Challenge: Extract the name, color, region of each bird into the \$subset array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

#### 📷 Part 1:

Progress: 100%

#### Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Heroku dev after a manual deploy)

```
// Start edits
// c1r26 07/13/2025
// For each member of birds, add a new array variable into the subset array
// Iterate through the contents of each member of birds and use the keys to identify desired data
// Copy only desired keys value pairs into the subset array

foreach ($birds as $index => $bird)
{
    $subset[$index] = [];
    foreach ($bird as $key => $value)
    {
        if ($key == "name" || $key == "color" || $key == "region")
        {
            $subset[$index][$key] = $value;
        }
    }
} <- #20-25 foreach ($bird as $key => $value)
} <- #17-26 foreach ($birds as $index => $bird)

// end edits
```

Subset code/comment



Completed Problem 1 for [ctrl6] (2025-07-14 13:05:46)

Subset output

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## Part 2:

Progress: 100%

### Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

#### URL #1

[https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public\\_html/M6/problem1.php](https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public_html/M6/problem1.php)



URL

<https://github.com/ColinRafferty7>



#### URL #2

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem1.php>



URL

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem1.php>



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## Part 3:

Progress: 100%

### Details:

Briefly explain `how` the code solves the challenge (note: this isn't the same as `what` the code does)

### Your Response:

The code solves the problem by utilizing the foreach loop. By iterating through the arrays that contain the data, I can get both the key and values for each member. Then I ran logic that only included the keys that were required for the output.

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## Section #2: ( 250 pts.) Problem 2 - Adding Properties

Progress: 100%

Task #1 ( 250 pts ) Edit the `processCars` function to add

# Task #1 (250 pts.) - Edit the processCars function to add properties

Progress: 100%

## Details:

- Only make edits where noted via provided comments
- Challenge 1: Add a new property called age that's set from today's year and the car's year
- Challenge 2: Add a new property called isClassic that's true/false based on \$classic\_age
- Step 1: sketch out plan using comments (includeucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

## Part 1:

Progress: 100%

## Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (withucid/date comment)
2. Full output of executing the program (visit the proper file on Heroku dev after a manual deploy)

```
// classic_age 2; // don't change this value
// Start edits
// 07/13/2025
// Set the current year to 2025
// For each member of cars, add a new array into processed cars
// Use an indexed for each loop to add each key value pair into the new array
// Use the newly added data to get the current age of the car and set the classic variable

$currentYear = 2025;

foreach ($cars as $index => $car)
{
    $processedCars[$index] = [];
    foreach ($car as $key => $value)
    {
        $processedCars[$index][$key] = $value;
    }
    $processedCars[$index]["age"] = ($currentYear - $processedCars[$index]["year"]);
    $processedCars[$index]["isClassic"] = ($processedCars[$index]["age"] > 2);
}
// End edits
```

Adding properties code/comment

| Car               | Year | Age | Is Classic |
|-------------------|------|-----|------------|
| 1965 Ford Mustang | 1965 | 60  | False      |
| 1965 Ford Mustang | 1965 | 60  | False      |
| 1965 Ford Mustang | 1965 | 60  | False      |
| 1965 Ford Mustang | 1965 | 60  | False      |

Adding properties output

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## Part 2:

**Details:**

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

**URL #1**

[https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public\\_html/M6/problem2.php](https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public_html/M6/problem2.php)



URL

<https://github.com/ColinRafferty7>**URL #2**

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem2.php>



URL

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem2.php>

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**Part 3:**

Progress: 100%

**Details:**

Briefly explain **how** the code solves the challenges (note: this isn't the same as **what** the code does)

**Your Response:**

I solved this problem in a very similar way as the first problem. I used the foreach loops to add all of the data from the original arrays. However, for the new values I included segments of code after all of the original data was copied. These segments calculated the age of the car using the entered year, as well as setting the isClassic flag to the output of the age being greater than or equal to 25.



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## Section #3: ( 250 pts.) Problem 3 - Join

Progress: 100%

**Task #1 ( 250 pts.) - Edit the `joinArrays` function to combine two arrays based on a common key**

Progress: 100%

**Details:**

- Only make edits where noted via provided comments
- Challenge: Combine the data in both arrays by the `userId` property



## Part 2:

Progress: 100%

### Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

#### URL #1

[https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public\\_html/M6/problem3.php](https://github.com/ColinRafferty7/ctr26-IT202-450-M6-Homework/public_html/M6/problem3.php)



URL

<https://github.com/ColinRafferty7>



#### URL #2

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem3.php>



URL

<https://ctr26-it202-450-prod-2ae7f60cad14.herokuapp.com/M6/problem3.php>



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## Part 3:

Progress: 100%

### Details:

Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

### Your Response:

The code solves the problem in a similar way to the second problem, but instead of adding only new key value pairs after the initial data, the program runs another foreach loop for the second array. This second loop skips the userId key since it is already included and then adds a new array for the value of activity.



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## Section #4: ( 250 pts.) Misc

Progress: 100%

### Task #1 ( 83.33 pts.) - Github Details

Progress: 100%

## Part 1:

Progress: 100%

### Details:

From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present

**M5 homework // 23**

ColinRafferty7 merged 8 commits into **main** from **PR: ctr26-IT202-450** 15 minutes ago

Conversation | Commits | Pull Checks | Plus Shared

ColinRafferty7 commented 15 minutes ago

ColinRafferty7 added 8 commits 15 minutes ago

- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450
- main: ctr26-IT202-450

ColinRafferty7 merged commit **main** into **main** 15 minutes ago

pull request successfully merged and closed

## Commit history

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## Part 2:

Progress: 100%

### Details:

Include the link to the Pull Request (should end in `/pull/#`)

### URL #1

<https://github.com/ColinRafferty7/ctr26-IT202-450>



URL

<https://github.com/ColinRafferty7>

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## Task #2 ( 83.33 pts.) - WakaTime - Activity

Progress: 100%

### Details:

- Visit the WakaTime.com Dashboard
- Click **Projects** and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

**Projects • ctr26-IT202-450**

**7 hrs 20 mins** over the **Last 7 Days** in **ctr26-IT202-450** under **all** branches.

## Waketime top

| Files                                            | Branches                              |
|--------------------------------------------------|---------------------------------------|
| 1 hr 27 mins public_html/project/login.php       | 3 hrs 3 mins Milestone1               |
| 1 hr 24 mins public_html/project/register.php    | 1 hr 31 mins M6-Homework              |
| 57 mins public_html/M6/problem3.php              | 45 mins Feat-M21-UserRegistration     |
| 56 mins lib/validations.php                      | 20 mins Feat-M51-UserProfile          |
| 53 mins public_html/M6/problem2.php              | 19 mins Feat-M51-HelperFunctions      |
| 20 mins partials/nav.php                         | 19 mins Feat-M51-UserRoles            |
| 19 mins public_html/M6/problem1.php              | 16 mins cleanup-m21-drying-out        |
| 16 mins public_html/project/sql/init_db.php      | 14 mins Feat-M51-BasicNav-Login       |
| 14 mins public_html/project/logout.php           | 14 mins ProjectSetup                  |
| 11 mins public_html/project/landing.php          | 8 mins Unknown                        |
| 10 mins public_html/project/profile.php          | 5 mins Feat-M51-FlashMessages         |
| 8 mins lib/user_helpers.php                      | 1 min dev                             |
| 6 mins lib/http-302redirect26/after-template.ssg | 54 secs Feat-M51-UserLoginEnhancement |
| 5 mins lib/helpers.php                           |                                       |
| 5 mins public_html/project/index.php             |                                       |
| 3 mins lib/safe_echo.php                         |                                       |
| 3 mins _q/002_after_table_users_username.sql     |                                       |
| 3 mins partials/flash.php                        |                                       |
| 3 mins public_html/test_db.php                   |                                       |
| 2 mins public_html/project/helpers.js            |                                       |
| 2 mins public_html/M6/base.php                   |                                       |

## Waketime bottom



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## Task #3 ( 83.33 pts.) - Reflection

Progress: 100%

### Task #1 ( 0.33 pts.) - What did you learn?

Progress: 100%

#### Details:

Briefly answer the question (at least a few decent sentences)

#### Your Response:

In this assignment, I developed my ability to manipulate arrays in php. I have a lot of experience with arrays in other languages, so most of the knowledge transferred over. It was good to get a feel for how the arrays function in php and I enjoy the logic for this language.



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### Task #2 ( 0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

#### Details:

Briefly answer the question (at least a few decent sentences)

#### Your Response:

The easiest part of this assignment was solving the problem once I had an idea of how the syntax and logic works in php. Arrays/lists are very similar in other languages and so I already had a good idea of how each problem could be solved.



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### ⇒ Task #3 ( 0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

#### Details:

Briefly answer the question (at least a few decent sentences)

#### Your Response:

The hardest part of this assignment was researching and testing the syntax and logic that php has for the arrays. It is slightly different from other languages, but after the first problem I got that hang of it. There is probably still many uses and nuances that I haven't used yet, but I have a good feel for the basics.



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